

Modular Abelian Varieties

MA4091 Seminar Matematika I

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Introduction

Background

- We define **elliptic curves** as follows:

Definition (Elliptic Curves)

An **elliptic curve** is the set of roots of the equation:

$$y^2z + a_1xyz + a_3yz^2 = x^3 + a_2x^2z + a_4xz^2 + a_6z^3$$

that lives in the projective plane.

- We say that an elliptic curve is defined over $\mathbb{Q}$ if all coefficients are in $\mathbb{Q}$ and we only consider solutions over $\mathbb{Q}$. (Silverman, 2009)
- We can define the structure of a group on these curves such that the resulting group is abelian.
- These sets of roots of a system of polynomials such that we can define an abelian group structure are called **abelian varieties**.

Background

- In 1995, Andrew Wiles and Richard Taylor proved (Wiles, 1995) (Taylor and Wiles, 1995):

Theorem (Fermat's Last Theorem)

Let $n > 2$ be an integer. Then all integer solutions of the equation

$$a^n + b^n = c^n$$

satisfies $abc = 0$.

- Along the way, they proved a special case of Taniyama-Shimura Conjecture:

Theorem (Taniyama-Shimura (Necessary to Prove FLT))

Every ("semi-stable") "elliptic curve" over $\mathbb{Q}$ arises from "modular forms".

Background

- Later on in 2001, Christophe Breuil, Brian Conrad, Fred Diamond, and Richard Taylor proved the full version of Taniyama-Shimura, now known as the **Modularity Theorem** (Breuil et al., 2001).
- In February 2025, George Boxer, Frank Calegari, Toby Gee, and Vincent Pilloni formulated and proved an analogue for the previous modularity theorem for abelian surfaces, which are 2-dimensional generalizations of elliptic curves (Boxer et al., 2025).
- For general abelian varieties (which are n -dimensional generalizations of elliptic curves) over an arbitrary ring, there is sadly still no news.
- These generalizations belong to the **Langlands program**.
- The proofs here requires modern scheme-theoretic algebraic geometry.

Aims for my Undergraduate Thesis

Thus, I propose the following aims:

- Understand the basic objects of algebraic geometry.
- Understand a more general version of the definition of elliptic curves.
- Compare the properties of classical elliptic curves as varieties with its more general version.
- Identify the role of scheme theory in the proof of modularity theorem.

Affine Schemes

Notation

Throughout this presentation, the following notations are standard:

- k : field
- $\bar{k}$: algebraic closure of k , or an algebraically closed field
- R : commutative ring with unity (or simply rings)
- $\mathfrak{a}$: ideal
- $\mathfrak{p}$: prime ideal
- $\mathfrak{m}$: maximal ideal
- $\subset$: proper inclusion

More standard notations to be introduced later along with their definitions.

Assumptions and Conventions

- As stated previously, all rings are assumed to be commutative with unity.
- The term "compact" means that every open cover has a finite subcover.
- The term "Hausdorff compact" means that it's compact and Hausdorff.
- The empty topological space is not connected.
- All ring homomorphisms $R \rightarrow S$ satisfy $1_R \mapsto 1_S$, $0_R \mapsto 0_S$.
- In this section, we closely follow Görtz and Wedhorn, 2020.

Varieties

- Classical algebraic geometry: the study of the set of roots of some polynomial.

Proposition

Define an **algebraic set** $Z(S)$ to be the set of all roots of the set of polynomials $S = \{f_1, \dots, f_m\}$ over k . We can define a topological space $\mathbb{A}_k^n$ of points in k^n where the closed sets are algebraic sets, called the **affine space**, and the topology is called the **Zariski topology**. If $S = \langle f_1, \dots, f_m \rangle$, then we write $Z(S) = Z(\langle f_1, \dots, f_m \rangle)$ as $Z(f_1, \dots, f_m)$.

- Do note that the set Z may be empty.

Definition

An **affine variety** over k , written as V , is an algebraic set $Z(\mathfrak{a})$, where $\mathfrak{a}$ is an ideal of $k[x_1, \dots, x_n]$.

Limitations with Varieties

- Problem: if we only consider the sets of points, we have a problem with multiplicities!
- Example: the algebraic sets $Z(x)$ and $Z(x^2)$, both subsets of $\mathbb{A}^1$, are **equal**, however $\langle x \rangle$ and $\langle x^2 \rangle$ are **different** ideals, where $\langle x \rangle$ is reduced and $\langle x^2 \rangle$ is not.
- Another problem: if k is not algebraically closed, then a polynomial over k might not have any roots in k !
- Example: $f(x) = x^2 + 16$ doesn't have **any** roots in $\mathbb{R}$, thus the empty variety and the algebraic set $Z(f)$ are equal.
- Thus we need a definition that could mitigate these problems, **especially** since later on we will work over $\mathbb{Q}$ or general "number fields".

Overview of Schemes

- Let's say we have a set X .
- There we can define a topological space (X, τ) .
- Then, for any open subset $U \subseteq X$, we can define a ring R_U , such that for any open subsets $U, V \subseteq X$, in $U \cap V$, we have "compatibility".
- In algebraic geometry, the set X will be related to some ring R .
- We do this, so that the algebraic information (example: multiplicities) are not lost.

Affine Schemes

Definition

Let R be a ring. The **prime spectrum**, or **spectrum**, of the ring R is defined by

$$\mathrm{Spec}(R) = \{\mathfrak{p} : \mathfrak{p} \subsetneq R, \mathfrak{p} \text{ is a prime ideal}\},$$

endowed with a topology where the closed sets are of the form

$$V(\mathfrak{a}) = \{\mathfrak{p} \subsetneq R : \mathfrak{a} \subseteq \mathfrak{p}\},$$

where $\mathfrak{a} \subseteq R$ is an ideal of R , called the **Zariski topology** of $\mathrm{Spec}(R)$.

- This topological space shall be used to generalize algebraic sets later on.
- We usually denote by $V(f)$ for $V(\langle f \rangle)$. The basis of the Zariski topology are the sets $D(f) = X \setminus V(f)$.

Affine Schemes

- Consider the following examples.
- In $\mathbb{Z}$, the prime numbers p generate prime ideals $\langle p \rangle$, and since $\mathbb{Z}$ is an integral domain, we have $\langle 0 \rangle$ is a prime ideal.
- Those are all the prime ideals of $\mathbb{Z}$, thus $\text{Spec } \mathbb{Z} = \{\langle p \rangle : p = 0 \text{ or } p \text{ prime}\}$.
- Thus the closed sets on $\text{Spec } \mathbb{Z}$ are of the form $V(n) = \{\langle p \rangle : p|n\}$, and the open sets are of the form $D(n) = \{\langle p \rangle : p \nmid n\}$.
- Another example is $R = \mathbb{C}[x]$.
- The prime ideals here are $\langle 0 \rangle$ and of the form $\langle x - \alpha \rangle = \{\beta(x - \alpha) : \beta \in \mathbb{C}[x]\}$.
- Thus $\text{Spec } R = \text{Spec } \mathbb{C}[x] = \{\langle p \rangle : p = 0 \text{ or } p = x - \alpha, \alpha \in \mathbb{C}\}$.
- The points in $\text{Spec } \mathbb{C}[x] \setminus \{\langle 0 \rangle\}$ corresponds to the points in $\mathbb{C}$, by $\langle x - \alpha \rangle \mapsto \alpha$.

Affine Schemes

- In $\text{Spec } \mathbb{C}[x]$, consider the ideal $\mathfrak{a} = \langle x^2 + 16 \rangle$.
- Notice that $x^2 + 16 = (x - 4i)(x + 4i)$ and $\mathbb{C}[x]$ is a principal ideal domain.
- This implies that $x^2 + 16 \in \langle x - 4i \rangle, \langle x + 4i \rangle$.
- These two ideals are the only prime ideals containing $x^2 + 16$, since if there exists an ideal $\langle f \rangle \neq \langle x \pm 4i \rangle$ such that $x^2 + 16 \in \langle f \rangle$, we must then have $f \mid (x - 4i)(x + 4i)$.
- Since $\mathbb{C}[x]$ is an integral domain, we have WLOG $f \mid (x - 4i)(x + 4i) \Rightarrow f \mid (x + 4i)$, thus $f = \alpha(x + 4i)$ for some $\alpha \in \mathbb{C}$.
- Then $\langle f \rangle = \langle x + 4i \rangle$, a contradiction.
- Thus we get $V(\mathfrak{a}) = \{\langle x - 4i \rangle, \langle x + 4i \rangle\} \mapsto \{\pm 4i\}$.

Affine Schemes

- Before we continue, we need a definition that would be extremely important.

Definition (Localization)

Let R be a ring and $S \subset R$ be a multiplicative subset, i.e. $1 \in S$ and if $u, v \in S$ then $uv \in S$. We define the **localization** of R with respect to S as

$$S^{-1}R = \left\{ \frac{a}{b} : a \in R, b \in S \right\} / \sim,$$

where $\sim$ is the equivalence relation defined as $\frac{a}{b} \sim \frac{a'}{b'}$ iff $s(ab' - a'b) = 0$ for some $s \in R$. If $0 \in S$, we define $S^{-1}R = \{0\}$.

- If $f \in R$, we define R_f as $\{f^i : i \geq 0\}^{-1}R$.

Affine Schemes

- Let $\mathfrak{p}, \mathfrak{q}$ be prime ideals of R .
- We shall consider localizations and quotients of R with respect to prime ideals.
- The prime ideals of $R_{\mathfrak{p}} = (R \setminus \mathfrak{p})^{-1}R$ are those that are contained in $\mathfrak{p}$.
- The prime ideals of $R/\mathfrak{q}$ are those that contains $\mathfrak{q}$.
- Thus if $\mathfrak{q} \subseteq \mathfrak{p}$, localizing with respect to $\mathfrak{p}$ and taking quotients with respect to $\mathfrak{q}$ yields a ring whose prime ideals are those that are contained in $\mathfrak{p}$ containing $\mathfrak{q}$.
- If $\mathfrak{p} = \mathfrak{q}$, then $\kappa(\mathfrak{p}) = R_{\mathfrak{p}}/\mathfrak{p}R_{\mathfrak{p}} \cong \text{Frac}(R/\mathfrak{p})$ is called the **residue field** of $\mathfrak{p}$.

Sheaves

- Consider the topological space $X = \mathbb{C}$.
- For any $U \subseteq X$ open, we define $\mathcal{F}(U) = \{f : U \rightarrow \mathbb{C} : f \text{ holomorphic}\}$.
- Notice that constant functions α are always holomorphic, thus $\mathcal{F}(U)$ are nonempty.
- Moreover, for $U = U_1 \cup U_2 \subseteq X$ where U, U_1, U_2 are open, if $f, g \in \mathcal{F}(U)$ such that $f|_{U_1} = g|_{U_2}$, we have $f = g$.
- From analytic continuation, if we have $f \in \mathcal{F}(U_1)$ and $g \in \mathcal{F}(U_2)$ such that $f|_{U_1 \cap U_2} = g|_{U_1 \cap U_2}$, then there exists $h \in \mathcal{F}(U)$ such that $h|_{U_1} = f$, $h|_{U_2} = g$.
- This construction can be generalized to arbitrary topological spaces X , which we need to define a scheme later.

Sheaves

Definition

Let X be a topological space. A **presheaf** $\mathcal{F}$ over X is a contravariant functor from (Ouv_X) (the category of open subsets of X) to a category $\mathcal{C}$. We call such presheaf **presheaf of ...** depending on $\mathcal{C}$.

- In concrete language, a **presheaf** $\mathcal{F}$ over X is for every open set $U \subseteq X$, a set/ring/group/etc $\mathcal{F}(U)$, and for every $U \subseteq V$ open, a mapping $\text{res}_U^V : \mathcal{F}(V) \rightarrow \mathcal{F}(U)$ called the **restriction map** such that:
 - 1 $\text{res}_U^U = \text{id}_{\mathcal{F}(U)}$ for every $U \subseteq X$ open,
 - 2 $\text{res}_U^W = \text{res}_U^V \circ \text{res}_V^W$ for $U \subseteq V \subseteq W$ open.
- Note that the restriction map is part of the data required to define a presheaf.

Sheaves

- The following notion is an analogue for "gluing" in varieties.

Definition

Let X be a topological space and $\mathcal{F}$ be a presheaf on X . We call $\mathcal{F}$ a **sheaf** over X if for $S \subseteq X$ open, $(S_i)_i$ an open cover of S ,

- ① If $s, s' \in \mathcal{F}(S)$ such that $s|_{S_i} = s'|_{S_i}$ for all i then $s = s'$,
- ② If there exists $s_i \in \mathcal{F}(S_i)$ for all i such that $s_i|_{S_i \cap S_j} = s_j|_{S_i \cap S_j}$, then there exist an $s \in \mathcal{F}(S)$ such that $s|_{S_i} = s_i$ for all i .

- For a topological space X , we use the notation $\mathcal{F}_X$ to denote a sheaf over X .

Sheaves

- Let X be a topological space, $U \subseteq X$ open, and $\mathcal{F}$ a (pre-)sheaf on X .
- We can obtain another (pre-)sheaf on U .

Definition

From the previous setting, define a (pre-)sheaf on U , $\mathcal{F}|_U$, by defining for all $V \subseteq U$ open $\mathcal{F}|_U(V) = \mathcal{F}(V)$. This (pre-)sheaf is called the **restriction** of $\mathcal{F}$ in U .

Sheaves

Definition

Let X be a topological space, $\mathcal{F}$ be a presheaf on X , and $x \in X$. A **stalk** of $\mathcal{F}$ in x is the set of equivalence classes of pairs (U, s) , where U is a neighborhood of x and $s \in \mathcal{F}(U)$. The pairs $(U_1, s_1), (U_2, s_2)$ are equivalent if there exists another open neighborhood $V \subseteq U_1 \cap U_2$ of x such that $s_1|_V = s_2|_V$.

- As an analogy, stalks can be thought of as the set of functions that are defined on a neighborhood of x .
- As an example, in $\mathbb{C}$ equipped with the sheaf of holomorphic functions $\mathcal{F}$, a stalk $\mathcal{F}_\alpha$ is defined as the pair (f, U) , where U is a neighborhood of α and f is a holomorphic function on U .
- The stalks $\mathcal{F}_\alpha$ is isomorphic to the ring of convergent power series in $z - \alpha$.

Affine Schemes

- By defining $\mathcal{O}_{\text{Spec } R}(D(f)) = R_f = \{f^i : i \geq 0\}^{-1}R$ for all basis $D(f)$, we get that $\text{Spec } R$ is a locally ringed space with the restriction maps defined, for $D(f) \subseteq D(g)$, to be homomorphisms $\rho_{f,g} : R_g \rightarrow R_f$, $a/g \mapsto (ah)/f^n$.
- We can see this by noticing that $D(f) \subseteq D(g)$ is equivalent to $V(g) \subseteq V(f)$, which implies $\langle f \rangle \subseteq \sqrt{\langle f \rangle} \subseteq \sqrt{\langle g \rangle}$, i.e. $f^n = hg$.

Definition

An **affine scheme** is a pair $(\text{Spec } R, \mathcal{O}_{\text{Spec } R})$, with $\text{Spec } R$ the spectrum of a ring R , and $\mathcal{O}_{\text{Spec } R}$ is a sheaf of rings, called the **structure sheaf**, defined as above.

- Thus if we have morphisms of affine schemes, we get that affine schemes form a category, which shall be denoted by (Aff) .
- We usually write $(\text{Spec } R, \mathcal{O}_{\text{Spec } R})$ as just $\text{Spec } R$, and for any $U \subseteq X$ open, $\mathcal{O}_X(U) = \Gamma(U, \mathcal{O}_X) = \left\{ (s_i)_i \in \prod_{U \subseteq D(f_i)} R_{f_i} : \text{for } D(f_j) \subseteq D(f_i), s_i|_j = s_j \right\}$.

Affine Schemes

- For R, S rings and $\varphi : R \rightarrow S$ a ring homomorphism, if $\mathfrak{p}$ is a prime ideal, we have $\varphi^{-1}(\mathfrak{p})$ is also a prime ideal.
- Thus we can define a morphism from $\operatorname{Spec} S$ to $\operatorname{Spec} R$ as $\mathfrak{p} \mapsto \varphi^{-1}(\mathfrak{p})$.
- Moreover, we can define a map of structure sheaves $f^b : \mathcal{O}_{\operatorname{Spec} R} \rightarrow f_* \mathcal{O}_{\operatorname{Spec} S}$ as, for $D(g) \subseteq Y$, $f^b_{D(g)} : \mathcal{O}_{\operatorname{Spec} R}(D(g)) \rightarrow \mathcal{O}_{\operatorname{Spec} S}(f^{-1}(D(g)))$.
- We can define another map of structure sheaves $f^\sharp : f^{-1} \mathcal{O}_{\operatorname{Spec} R} \rightarrow \mathcal{O}_{\operatorname{Spec} S}$.

Proposition

A morphism of affine schemes $(f, f^b) : (\operatorname{Spec} S, \mathcal{O}_{\operatorname{Spec} S}) \rightarrow (\operatorname{Spec} R, \mathcal{O}_{\operatorname{Spec} R})$ consists of a continuous map f arising from a homomorphism of rings $\varphi : R \rightarrow S$ and a map of structure sheaves $f^b : \mathcal{O}_{\operatorname{Spec} R} \rightarrow f_* \mathcal{O}_{\operatorname{Spec} S}$.

Affine Schemes

- We consider the following examples.
- All previous examples of Spec are affine schemes.
- If R is a ring, then $\mathbb{A}_R^n = \operatorname{Spec} R[X_1, \dots, X_n]$ is called the **affine space of relative dimension n over R** .
- For $n = 2$, we have $\mathbb{A}_R^2 = \operatorname{Spec} R[X_1, X_2]$.
- We see that the closed sets here are of the form $V(\mathfrak{a})$ for some ideal $\mathfrak{a}$.
- Thus the open sets are of the form $D(\mathfrak{a}) = (\operatorname{Spec} R[X_1, X_2]) \setminus V(\mathfrak{a})$.

Affine Schemes

- In $\mathbb{A}_{\mathbb{R}}^1$, consider the ideal $\mathfrak{a} = \langle x^2 + 16 \rangle$.
- This generates the empty variety, but generates the scheme $X := \operatorname{Spec} \mathbb{R}[x]/\langle x^2 + 16 \rangle$.
- The underlying topological space is nonempty, since $\langle 0 \rangle$ is an element.
- Moreover, the polynomial $x^2 + 16$ is irreducible over $\mathbb{R}$, thus the ideal $\langle x^2 + 16 \rangle$ is maximal.
- Thus $\operatorname{Spec} \mathbb{R}[x]/\langle x^2 + 16 \rangle = \{\langle 0 \rangle\}$.
- Moreover, the structure sheaf is

$$\mathcal{O}_x(D(1)) = \{1\}^{-1} \mathbb{R}[x]/\langle x^2 + 16 \rangle \cong \mathbb{R}[x]/\langle x^2 + 16 \rangle \cong \mathbb{R}[4i] \cong \mathbb{C}.$$

Varieties as Schemes

- Now we shall describe how varieties are related to schemes.
- Recall that affine varieties are defined as the zero set of some ideal in $k[x_1, \dots, x_n]$.
- Let $V \subseteq \mathbb{A}^n$ be an affine variety defined by $V(\mathfrak{a})$.
- Then we can define a ring $R = k[x_1, \dots, x_n]/\mathfrak{a}$, and an affine scheme $\text{Spec } R$.
- Let $\text{Specmax } R = \{\mathfrak{m} \subsetneq R : \mathfrak{m} \text{ is a maximal ideal of } R\}$.
- The space $\text{Specmax } R$ consists of the points $\langle x_1 - \alpha_1, \dots, x_n - \alpha_n \rangle$, which by the mapping $\langle x_i - \alpha_i \rangle_i \mapsto (\alpha_i)_i$, we have $\text{Spec}(R) \setminus \{\langle 0 \rangle\} \cong V(\mathfrak{a})$ as a variety.
- Thus we get that all affine varieties are related to schemes.

Varieties as Schemes

- However, there are distinct schemes that generate the same variety.
- For example, the scheme $\text{Spec } \mathbb{C}[x]/\langle x^2 \rangle$ is related to the algebraic set $Z(x^2) = Z(x)$, since $\langle x^2 \rangle$ is not a prime ideal.
- This is actually great: the only open set of $\text{Spec } \mathbb{C}[x]/\langle x^2 \rangle$ is $\{\langle x \rangle\}$.
- Thus we see that the structure sheaf of $\text{Spec } \mathbb{C}[x]/\langle x^2 \rangle$ is just

$$\mathcal{O}_{\text{Spec } \mathbb{C}[x]/\langle x^2 \rangle}(D(1)) = \{1\}^{-1} \mathbb{C}[x]/\langle x^2 \rangle = \mathbb{C}[x]/\langle x^2 \rangle.$$

- However, the structure sheaf of a related scheme, $\text{Spec } \mathbb{C}[x]/\langle x \rangle$, is just

$$\mathcal{O}_{\text{Spec } \mathbb{C}[x]/\langle x \rangle}(D(1)) = \{1\}^{-1} \mathbb{C}[x]/\langle x \rangle = \mathbb{C}[x]/\langle x \rangle \neq \mathbb{C}[x]/\langle x^2 \rangle.$$

Varieties are Schemes

- By the following theorem, any information from varieties are not lost when we transfer the language to schemes.
- To do this, let $Z(\mathfrak{a})$ be a variety over $\mathbb{A}_R^n$.
- We can associate an affine scheme $(X, \mathcal{O}_X)$ with $Z(\mathfrak{a})$ by letting $X = \text{Spec } R[x_1, \dots, x_n]/\mathfrak{a}$, thus yielding:

Theorem (Varieties are Schemes)

Let $Z(\mathfrak{a})$ be a variety. Then there exists an affine scheme related to $Z(\mathfrak{a})$ defined as above.

- Moreover, we have the following equivalence.

Theorem

The category of affine schemes are equivalent to the category of rings.

Schemes

Schemes

- Using affine schemes, we can now properly define schemes.

Definition

A **scheme** is a locally ringed space $(X, \mathcal{O}_X)$ with an open cover $X = \bigcup_i U_i$, such that all locally ringed spaces $(U_i, \mathcal{O}_{X|U_i})$ are affine schemes.

A **morphism of schemes** is a morphism of locally ringed spaces.

- We obtain a category of schemes (Sch). We denote a scheme $(X, \mathcal{O}_X)$ as simply X whenever the sheaf is not considered.

Schemes

- Since a scheme requires a topological space, we can analogously define "openness" of a scheme.

Definition (Also a Proposition)

Let X be a scheme.

- ① If $U \subseteq X$ is open, then $(U, \mathcal{O}_{X|U})$ is a(n affine) scheme, called an **(affine) open subscheme** of X .
- ② The underlying topological spaces of affine open subschemes form a basis of the topology of X .

Proof Sketch.

If X is a scheme, then it is covered by affine schemes $(U_i)_i$ that are also open. If $U \subseteq X$ is open, then $U \cap U_i$ is also open. □

Schemes

- If we have a scheme S , we can define another category related to schemes. This is the basis for Grothendieck's "relative point of view".

Definition

The category (Sch/S) of **schemes over S** or **S -schemes** is the category whose objects are the morphisms of schemes $X \rightarrow S$, and whose morphisms $\text{Hom}(X \rightarrow S, Y \rightarrow S)$ are the morphisms $X \rightarrow Y$ such that the following diagram commutes:

$$\begin{array}{ccc} X & \xrightarrow{\quad} & Y \\ & \searrow & \swarrow \\ & S & \end{array}$$

The scheme S is called the **base scheme**, and the morphism $X \rightarrow S$ is called the **structural morphism** of the S -scheme X .

Schemes

- As a concrete example, consider the k -scheme

$$X = \operatorname{Spec} k[x_1, x_2] / \langle x_1^2 + x_2^2 - 1 \rangle \rightarrow \operatorname{Spec} k.$$

- Notice that every point of $\operatorname{Spec} k[x_1, x_2] / \langle x_1^2 + x_2^2 - 1 \rangle$ is associated to a prime ideal $\mathfrak{p}$ of $k[x_1, x_2] / \langle x_1^2 + x_2^2 - 1 \rangle$.
- Thus every point of X is of the form $\mathfrak{p} \mapsto \langle 0 \rangle$, which for the sake of laziness we shall write as $\mathfrak{p}$.
- Since every element of $k[x_1, x_2] / \langle x_1^2 + x_2^2 - 1 \rangle$ is of the form $f(x_1, x_2) \bmod x_1^2 + x_2^2 - 1$, every prime ideal of $k[x_1, x_2] / \langle x_1^2 + x_2^2 - 1 \rangle$ is of the form $\mathfrak{p} \bmod \langle x_1^2 + x_2^2 - 1 \rangle$.
- These points corresponds to the points in the circle $x_1^2 + x_2^2 - 1 = 0$ intersecting the generators of $\mathfrak{p}$ (finite since $k[x_1, x_2] / \langle x_1^2 + x_2^2 - 1 \rangle$ is Noetherian).

Schemes

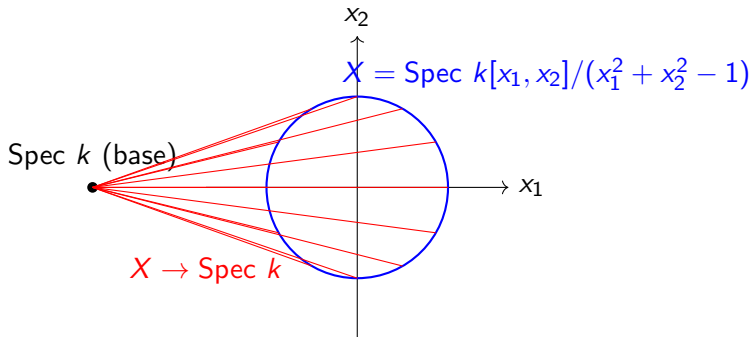


Figure 1: The scheme $\operatorname{Spec} k[x_1, x_2]/(x_1^2 + x_2^2 - 1)$ over $\operatorname{Spec} k$

Scheme

- Classically, the projective plane $\mathbb{P}_k^2$ is defined as the set $\mathbb{P}_k^2 = \{[a_1 : a_2 : a_3] : [a_1 : a_2 : a_3] = [\lambda a_1 : \lambda a_2 : \lambda a_3] \ \forall 0 \neq \lambda \in k\}$.
- Consider three affine planes $\mathbb{A}_1^2 = \text{Spec } R[x/z, y/z]$, $\mathbb{A}_2^2 = \text{Spec } R[y/x, z/x]$, and $\mathbb{A}_3^2 = \text{Spec } R[x/y, z/y]$.
- We define P^2 as the union of the three affine planes.
- We "glue" these three affine planes together by defining:

$$R[x/z, y/z] \rightarrow R[y/x, z/x], \quad y \mapsto y, \quad \frac{x}{z} \mapsto \frac{z}{x},$$

and we define the other two morphisms similarly.

- Thus we get the **projective plane** $\mathbb{P}_R^2$.
- This construction can be generalized, yielding the **projective space** $\mathbb{P}_R^n$.

Schemes

- We can "glue" multiple morphisms of schemes in order to define a new morphism of schemes.

Proposition

Let X, Y be schemes with open cover $X = \bigcup_i U_i$. A family of morphisms $\varphi_i : U_i \rightarrow Y$ glues to a morphism $X \rightarrow Y$ if and only if the morphisms coincide on intersections $U_i \cap U_j$, and the resulting morphism $X \rightarrow Y$ is uniquely determined.

Proof.

$\Rightarrow$ If we assume otherwise, on an intersection $U_i \cap U_j$ we have $\varphi|_{U_i \cap U_j} = \varphi_i|_{U_i \cap U_j} \neq \varphi_j|_{U_i \cap U_j} = \varphi|_{U_i \cap U_j}$, contradiction.

$\Leftarrow$ We define the morphism directly. □

Schemes

- Just like morphism of schemes, we can "glue" multiple schemes in order to define a new scheme.

Definition

A **gluing datum of schemes** consists of the following:

- ① an index set I ,
- ② for all $i \in I$ a scheme U_i ,
- ③ for all $i, j \in I$ an open subscheme $U_{ij} \subseteq U_i$,
- ④ for all $i, j \in I$ an isomorphism of schemes $\varphi_{ji} : U_{ij} \rightarrow U_{ji}$,

such that $U_{ii} = U_i$ for all $i \in I$ and the cocycle condition holds: $\varphi_{kj} \circ \varphi_{ji} = \varphi_{ki}$ on $U_{ij} \cap U_{ik}$, $i, j, k \in I$.

Theorem (Gluing Schemes)

Gluing datum of schemes yields a scheme X , unique up to unique isomorphism.

Schemes

- Let X be a scheme.
- Just like subsets are to sets, we have the following notions.

Definition

- ① A **closed subscheme** Y of X is a closed subset $Y \subseteq X$ with an ideal sheaf $\mathcal{I} \subseteq \mathcal{O}_X$ such that $Y = \{x \in X : (\mathcal{O}_X/\mathcal{I})_x \neq 0\}$ and $(Y, (\mathcal{O}/\mathcal{I})|_Y)$ is a scheme.
- ② A **subscheme** Y of X is a scheme Y such that $Y \subseteq X$ is locally closed, and Y is a closed subscheme of the open subscheme $U \subseteq X$, where U is the largest open subset of X which contains Y and Y is closed there.
- ③ An **immersion** $\iota : Y \rightarrow X$ is a morphism of schemes whose underlying continuous map is a homeomorphism to a locally closed subset of X , and for all $y \in Y$ the ring homomorphism $\iota_y^\# : \mathcal{O}_{X, \iota(y)} \rightarrow \mathcal{O}_{Y, y}$ is surjective.

Fiber Products

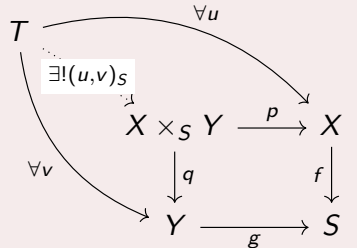
Fiber Products

- To define operations on an abelian variety properly, we need the notion of a "product".
- Since we shall mostly work with S -schemes, we can define operations easily.
- However, we still need a definition that translates the language of S -schemes to schemes.
- More importantly, products of schemes are not products of sets: the underlying set of $\mathbb{A}_k^2$ is not equal to the underlying set of $\mathbb{A}_k^1 \times \mathbb{A}_k^1$, even though as schemes both of them are isomorphic.

Fiber Products

Definition (Fiber Product)

Let X, Y be arbitrary schemes and $f : X \rightarrow S$ and $g : Y \rightarrow S$ be morphisms. The **fiber product** of X and Y consists of a scheme $X \times_S Y$ and morphisms $p : X \times_S Y \rightarrow X$, $q : X \times_S Y \rightarrow Y$ satisfying the following universal property:



Fiber Products

- Later on, when we consider group schemes, we need a new functor representing a scheme X .
- Coincidentally, the functor recovers the geometric intuition of varieties to the language of schemes by defining the following functor:

$$h_X : (\text{Sch})^{\text{opp}} \rightarrow (\text{Sets})$$

$$T \mapsto h_X(T) = X(T) = \text{Hom}_{(\text{Sch})}(T, X)$$

$$(f : T' \rightarrow T) \mapsto (\text{Hom}(T, X) \rightarrow \text{Hom}(T', X), g \mapsto g \circ f)$$

- We think of the functor h_X as "substituting" T to " X ".
- The previous definition also works if we replace (Sch) with (Sch/S) for any scheme S , for which we write $h_{X_S}(T) = X_S(T)$ instead.

Fiber Products

- Let $(X \times_S Y, p, q)$ be a fiber product of X and Y .
- By the universal property, for all S -schemes $T \rightarrow S$, the following map is a bijection:

$$\begin{aligned} \mathrm{Hom}_{(\mathrm{Sch}/S)}(T, X \times_S Y) &\rightarrow \mathrm{Hom}_{(\mathrm{Sch}/S)}(T, X) \times \mathrm{Hom}_{(\mathrm{Sch}/S)}(T, Y) \\ w &\mapsto (p \circ w, q \circ w). \end{aligned}$$

- Thus we see that the set of morphisms $\mathrm{Hom}_{(\mathrm{Sch}/S)}(T, X \times_S Y)$ of S -schemes is the same as the cartesian product $\mathrm{Hom}_{(\mathrm{Sch}/S)}(T, X) \times \mathrm{Hom}_{(\mathrm{Sch}/S)}(T, Y)$ of the set of morphisms of schemes.

Fiber Products

- As a consequence of the following lemma, fiber products of schemes always exists.
- To see this, remember that by definition a scheme X is covered by affine schemes.

Lemma

Let $A \leftarrow R \rightarrow B$ be ring homomorphisms, $S = \operatorname{Spec}(R)$, $X = \operatorname{Spec}(A)$, and $Y = \operatorname{Spec}(B)$. Set $Z = \operatorname{Spec}(A \otimes_R B)$ and let $p : Z \rightarrow X$ and $q : Z \rightarrow Y$ be the morphisms of schemes corresponding to the ring homomorphisms

$$\alpha : A \rightarrow A \otimes_R B, \quad a \mapsto a \otimes 1,$$

$$\beta : B \rightarrow A \otimes_R B, \quad b \mapsto 1 \otimes b.$$

Then (Z, p, q) is a fiber product of X and Y over S in (Aff) .

Fiber Products

Here we consider the following examples.

- Let $X = \operatorname{Spec} k[T_1, \dots, T_n]$ and $Y = \operatorname{Spec} k[T_{n+1}, \dots, T_m]$ be k -schemes, where $m > n$. Then $X \times_k Y = \operatorname{Spec} k[T_1, \dots, T_m]$. In other words, the fiber product of $\mathbb{A}_k^n$ and $\mathbb{A}_k^{m-n}$ is $\mathbb{A}_k^m$.
- Let $X = \operatorname{Spec} k[x_1, x_2]/\langle x_1^2 + x_2^2 - 1 \rangle$ and $Y = \operatorname{Spec} k[y_1]$ be k -schemes. Then

$$\begin{aligned} X \times_k Y &= \operatorname{Spec} k[x_1, x_2]/\langle x_1^2 + x_2^2 - 1 \rangle \times_k \operatorname{Spec} k[y_1] \\ &\cong \operatorname{Spec}(k[x_1, x_2]/\langle x_1^2 + x_2^2 - 1 \rangle \otimes_k k[y_1]) \\ &\cong \operatorname{Spec}(k[x_1, x_2, y_1]/\langle x_1^2 + x_2^2 - 1 \rangle). \end{aligned}$$

This is a cylinder of radius 1.

Fiber Products

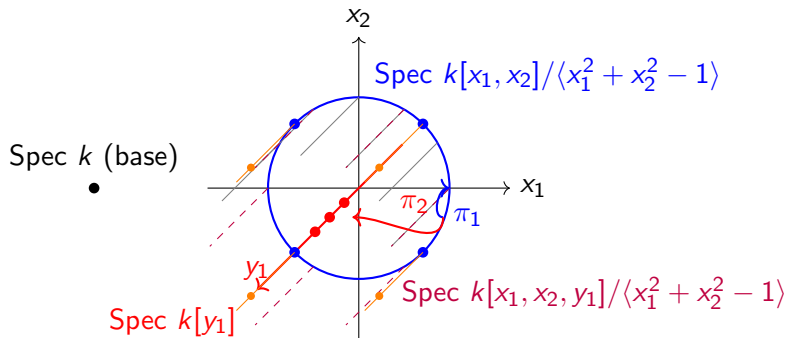


Figure 2: Fiber product of $\text{Spec } k[x_1, x_2]/\langle x_1^2 + x_2^2 - 1 \rangle$ and $\text{Spec } k[y_1]$ over $\text{Spec } k$

Fiber Products

- As a nontrivial example, we consider $X \times_S Y$ where $S = \operatorname{Spec} k[z_1]$, $X = \operatorname{Spec} k[x_1, x_2]/\langle x_1^2 + x_2^2 - 1 \rangle$, and $Y = \operatorname{Spec} k[y_1]$.
- Let $f : X \rightarrow S$ and $g : Y \rightarrow S$ be the structural morphisms of X, Y , with associated ring homomorphisms φ, ψ respectively.
- Then we have

$$\begin{aligned} X \times_S Y &= \operatorname{Spec} k[x_1, x_2]/\langle x_1^2 + x_2^2 - 1 \rangle \times_{\operatorname{Spec} k[z_1]} \operatorname{Spec} k[y_1] \\ &\cong \operatorname{Spec}(k[x_1, x_2]/\langle x_1^2 + x_2^2 - 1 \rangle \otimes_{k[z_1]} k[y_1]). \end{aligned}$$

- By considering $k[x_1, x_2]/\langle x_1^2 + x_2^2 - 1 \rangle$ and $k[y_1]$ as $k[z_1]$ -modules by the ring homomorphisms $f_S^\flat$ and $g_S^\flat$, we have

$$X \times_S Y \cong \operatorname{Spec}(k[x_1, x_2, y_1]/(\langle x_1^2 + x_2^2 - 1 \rangle + \langle \varphi(z_1) - \psi(z_1) \rangle)).$$

Fiber Products

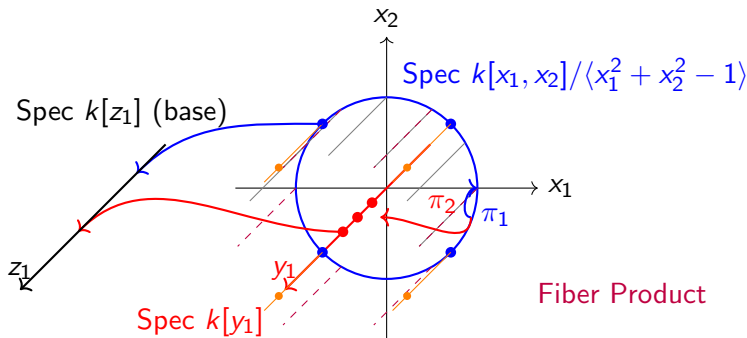


Figure 3: Fiber product of $\text{Spec } k[x_1, x_2]/\langle x_1^2 + x_2^2 - 1 \rangle$ and $\text{Spec } k[y_1]$ over $\text{Spec } k[z_1]$

Fiber Products

- Now we shall define the notion of **fiber**.

Definition

Let $f : X \rightarrow S$ be a morphism of schemes and $s \in S$. The topological fiber $f^{-1}(s)$ can be endowed with the structure of a scheme, by letting $\text{Spec } \kappa(s) \rightarrow S$ be the canonical morphism. Then we call

$$X_s = X \otimes_S \kappa(s) = X \times_S \text{Spec } \kappa(s) = f^{-1}(s)$$

the **fiber** of f in s .

Fiber Products

- As an example, let $X = \operatorname{Spec} \mathbb{C}[x]$ and $S = \operatorname{Spec} \mathbb{R}[x]$.
- Define a morphism of schemes $f : X \rightarrow S$ as the map generated by $\varphi : \mathbb{R}[x] \rightarrow \mathbb{C}[x]$, $\varphi(p) = p(x - i)$.
- Then the fiber of $\langle x - \alpha \rangle$ is $\langle x - \alpha - i \rangle$.
- At the same time, the residue field $\kappa(s)$ for any nonzero points $s = \langle x - \alpha \rangle$ are $\kappa(s) = \mathbb{R}[x]_{\langle x - \alpha \rangle} \cong \mathbb{R}$.

$$X \times_S \operatorname{Spec} \kappa(s) = \operatorname{Spec}(\mathbb{C}[x] \otimes_{\mathbb{R}[x]} \mathbb{R}) \cong \operatorname{Spec} \mathbb{C}.$$

Fiber Products

- Let X be an S -scheme and T be a scheme.
- Sometimes we need to change the basis of X from S to T , which we shall do later when we talk about elliptic curves.
- To do this, we define a morphism $u : T \rightarrow S$ and the fiber product $X \times_S T$.
- The scheme $X \times_S T$ is a T -scheme via the second projection, called the **base change** of X by u .
- This is one of the main advantages of using S -schemes: we can change the base to be a "tame" scheme, such as $\operatorname{Spec} \mathbb{Q}$ (or to mostly any scheme, really).
- The other main advantage: by considering an S -scheme, we are actually considering a family of schemes, by just changing the base.

Fiber Products

- As an example, we shall change the basis of $\operatorname{Spec} \mathbb{R}[x]/\langle x^2 + 16 \rangle$ from $\mathbb{R}$ to $\mathbb{C}$.
- Let $u : \operatorname{Spec} \mathbb{C} \rightarrow \operatorname{Spec} \mathbb{R}$ be the canonical morphism, $\operatorname{Spec} \mathbb{R}[x]/\langle x^2 + 16 \rangle \times_{\operatorname{Spec} \mathbb{R}} \operatorname{Spec} \mathbb{C}$ be the fiber product.
- By this morphism, we have

$$\begin{aligned} \operatorname{Spec} \mathbb{R}[x]/\langle x^2 + 16 \rangle \times_{\operatorname{Spec} \mathbb{R}} \operatorname{Spec} \mathbb{C} &\cong \operatorname{Spec} \mathbb{R}[x]/\langle x^2 + 16 \rangle \otimes_{\mathbb{R}} \mathbb{C} \\ &\cong \operatorname{Spec} \mathbb{C}[x]/\langle x^2 + 16 \rangle \cong \operatorname{Spec} \mathbb{C}[x]/\langle x - 4i \rangle \cup \operatorname{Spec} \mathbb{C}[x]/\langle x + 4i \rangle. \end{aligned}$$

- The final scheme is the disjoint union of two points plus a generic point.

Group Scheme

- The correct notion of a **group scheme** over a scheme uses fiber products.
- Let (Grp) be the category of groups, $V : (\text{Grp}) \rightarrow (\text{Sets})$ be the forgetful functor, S be a scheme and G be an S -scheme.
- The following structures are equivalent by the Yoneda lemma:
 - ① A factorization of the functor $G_S : (\text{Sch}/S)^{\text{opp}} \rightarrow (\text{Sets})$ through V ,
 - ② For all S -schemes T the structure of a group $G_S(T)$ which is functorial in T ,
 - ③ Three S -morphisms $m : G \times_S G \rightarrow G$ (multiplication), $i : G \rightarrow G$ (inversion), and $e : S \rightarrow G$ (unit) such that the diagrams in the next slide all commute.

An S -scheme together with (one of these) additional structures is called a **group scheme** over S .

Group Scheme

$$\begin{array}{ccccc}
 G \times_S G \times_S G & \xrightarrow{m \times \text{id}_G} & G \times_S G & & G \times_S S & \xrightarrow{(\text{id}_G, e)_S} & G \times_S G & & G & \xrightarrow{(\text{id}_G, i)_S} & G \times_S G \\
 \text{id}_G \times m \downarrow & & \downarrow m & & \downarrow = & & \downarrow m & & \downarrow & & \downarrow m \\
 G \times_S G & \xrightarrow{m} & G & & G & \xrightarrow{\text{id}_G} & G & & S & \xrightarrow{e} & G
 \end{array}$$

Figure 4: Associativity, Unit, and Inversion Diagrams.

Group Scheme

- In category theory, a family of morphisms $\alpha(S) : F(S) \rightarrow G(S)$ for all objects S is called **functorial** in S , or a **morphism of functors**, if the following diagram commutes:

$$\begin{array}{ccc}
 F(T) & \xrightarrow{\alpha(T)} & G(T) \\
 F(f) \downarrow & & \downarrow G(f) \\
 F(S) & \xrightarrow{\alpha(S)} & G(S)
 \end{array}$$

Figure 5: Functorial in S

Yoneda Lemma

- Here we state and prove the Yoneda Lemma, which will be used to prove that the three data in the definition of group schemes are equivalent.

Lemma (Yoneda)

Let $\hat{\mathcal{C}}$ be the category of functors $\mathcal{C}^{\text{opp}} \rightarrow (\text{Sets})$, and F is an object in $\hat{\mathcal{C}}$. The map $\text{Hom}_{\hat{\mathcal{C}}}(h_X, F) \rightarrow F(X), \alpha \mapsto \alpha_X(\text{id}_X)$ is bijective and functorial in X .

- Here, $\text{Hom}_{\hat{\mathcal{C}}}(h_X, F)$ consists of morphisms of functors $\alpha : h_X \rightarrow F$.
- From the Yoneda Lemma, we have the following corollary.

Corollary

If $F = h_Y$, we have $\text{Hom}_{\hat{\mathcal{C}}}(h_X, h_Y) \cong \text{Hom}_{\mathcal{C}}(X, Y)$.

- Using Yoneda Lemma notations, we have $h_{X \times_S Y/S}(T) \cong h_{X/S}(T) \times h_{Y/S}(T)$.

Yoneda Lemma

- First we prove that $F(X)$ is nonempty, by proving $\alpha_X(\text{id}_X) \in F(X)$ for all X .
- Since α is a morphism of functors, there exists a map $\alpha_X : h_X(X) = \text{Hom}_{\mathcal{C}}(X, X) \rightarrow F(X)$.
- Since $\text{id}_X \in \text{Hom}_{\mathcal{C}}(X, X)$, we have $\alpha_X(\text{id}_X) \in F(X)$.
- Now let $\xi \in F(X)$, and define the mapping $\alpha_\xi(Y) : h_X(Y) \rightarrow F(Y)$ as $f \mapsto F(f)(\xi)$ for $f \in h_X(Y)$.
- We know that α_Y is functorial in Y .

Yoneda Lemma

- We prove that $\xi \mapsto \alpha_\xi$ is the inverse of $\alpha \mapsto \alpha(X)(\text{id}_X)$.
- If $\xi \in F(X)$ and $\alpha \in \text{Hom}_{\hat{\mathcal{C}}}(h_X, F)$, then

$$\begin{aligned}\xi \mapsto \alpha_\xi &= F(-)(\xi) \mapsto \alpha_\xi(X)(\text{id}_X) = F(\text{id}_X)(\xi) = \xi, \\ \alpha \mapsto \alpha_X(\text{id}_X) &\mapsto \alpha_{\alpha(X)(\text{id}_X)} = \alpha_{F(\text{id}_X)(-)} = \alpha.\end{aligned}$$

- To prove that it is functorial, let $\alpha \in \text{Hom}_{\hat{\mathcal{C}}}(h_X, F)$. Then for any morphism $f : Y \rightarrow X$,

$$\begin{aligned}\alpha \mapsto \alpha_X(\text{id}_X) &\mapsto F(f) \circ \alpha_X(\text{id}_X) = \alpha_Y \circ h_X(f)(\text{id}_X) = \alpha_Y(f), \\ \alpha \mapsto \alpha \circ h_f &\mapsto \alpha_Y \circ h_f(Y)(\text{id}_Y) = \alpha_Y(f).\end{aligned}$$

- Here, $h_f(S) : h_X(S) \rightarrow h_Y(S), g \mapsto f \circ g$.

Group Schemes

- As a corollary, we have the main claim:

Theorem (Main Theorem)

Let S be a scheme, G be an S -scheme, and $V : (\text{Grp}) \rightarrow (\text{Sets})$ be the forgetful functor. The following structures are equivalent:

- ① *A factorization of the functor $G_S : (\text{Sch}/S)^{\text{opp}} \rightarrow (\text{Sets})$ through V ,*
- ② *For all S -schemes T the structure of a group $G_S(T)$ which is functorial in T ,*
- ③ *Three S -morphisms $m : G \times_S G \rightarrow G$ (multiplication), $i : G \rightarrow G$ (inversion), and $e : S \rightarrow G$ (unit) such that the diagrams in slide 52 all commute.*

- Thus G/S is a group scheme if and only if $G_S(T)$ is a group for all S -schemes T .
- Note that the equivalence $(1) \Leftrightarrow (2)$ is trivial.

Next Steps

Next Steps

- As a prerequisite to elliptic curves, we need the notions of **divisor** and **line bundles**, which I need to learn.
- After learning that, I need to learn a generalization of the definition of elliptic curves and give some nontrivial examples.
- Then I shall compare the properties of classical elliptic curves, defined as vanishing schemes, with the generalization that I learned.
- Finally, I shall identify the role of schemes in the proof of the modularity theorem.

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Thanks for your attention.
Any questions?